

Student Platform – 12-Month Cloud Costing

ILLUSTRATIVE

7,000 daily active users · document upload & review · India-region cloud hosting (AWS ap-south-1 benchmark)

₹11.82 L

Recommended sanction figure —
1-yr commit + 15% growth
headroom

₹10.28 L

Bare infrastructure cost, 12
months, no headroom

14.6 TB

Document storage on disk by
month 12

35%

Of the bill is bandwidth, not
servers — the real cost driver

A · WHAT WE'RE SIZING FOR

Metric	Value	Basis
Daily active users	7,000	As given — students interacting with the platform each day
Peak concurrent users	~1,000	Assumed 15% of DAU active in the same hour — school-day usage pattern, not spread evenly across 24h
Uploads per student per day	1.5 documents	Assumption — certificates, ID proof, assignments, photos. Confirm against real usage once live.
Average file size	3 MB	Assumption — mixed scanned PDFs and photos. A phone photo alone often runs 2–5 MB.
New raw uploads	30.8 GB / day	7,000 × 1.5 × 3 MB
Re-views per document	3×	Assumption — verification, re-checking, coach/admin review each pull the file again
Total daily bandwidth	~104 GB / day	Document re-views (92 GB) + general app/API traffic (12 GB)

B · MONTHLY INFRASTRUCTURE, AT STEADY STATE

Component	₹/month	Basis
Compute & database		
App servers — 2× (2 vCPU/8GB), always-on	11,680	HA pair, no single point of failure
Load balancer	2,110	Routes to the app pair, auto-scales on spikes
Database — PostgreSQL, Single-AZ	17,730	User records, upload metadata, audit trail
Database storage — 300 GB SSD	2,880	Metadata only — files live in object storage, not the DB
App disk — 150 GB SSD	1,440	OS, logs, temp space
Security & operations		
Web application firewall	1,700	Blocks scraping, injection, brute-force login attempts
Upload virus/malware scan	3,840	Every uploaded file scanned before it's stored
Monitoring & alerting	2,880	Uptime, error rates, capacity warnings
DNS, certificates, misc.	480	Domain routing, TLS
Storage & bandwidth — grows monthly		
Document storage — month 1 → month 12	2,550 → 30,640	Cumulative, +30% buffer for versioning/backup
Bandwidth — downloads & app traffic	33,170	Flat once usage stabilises, ~3.2 TB/month
Total, by month 12	108,550	Peak monthly run-rate, not the year-1 average

Category	₹ / 12 mo	Share
Bandwidth — document downloads	3,98,080	38.7%
Compute & database (1-yr commit)	2,47,050	24.0%
Document storage (growing)	1,99,130	19.4%
Security, monitoring, misc.	1,06,750	10.4%
App/DB disk, load balancer	77,180	7.5%
Total, 1-yr-commit equivalent	10,28,190	100%

Bandwidth beats compute. Serving documents back to students and reviewers costs more than running the servers that handle them. The efficient lever here is caching and view-limits, not a bigger server.

EFFICIENCY CHOICES ALREADY BUILT IN

- **Uploads go student → storage directly** (pre-signed URLs), bypassing the app server entirely — the biggest single cost avoidance, since AWS does not charge for data coming in.
- **Downloads are served through a CDN**, not straight off the server — ~16% cheaper per GB than direct egress, and faster for students in remote districts.
- **A 1-year Reserved commitment on compute + database** cuts that line by ~30% against paying by the hour — worth locking in since this is already a 12-month plan.
- **Single-AZ database** by default. Multi-AZ (automatic failover) adds **₹2.13 L/year** — a real option if downtime during a submission deadline is unacceptable, listed here as an add-on rather than assumed.

NOT INCLUDED HERE

- Document processing beyond storage — OCR, auto-verification, or AI review of uploaded certificates, if wanted later.
- SMS/WhatsApp notifications to students on upload status.
- A staging/test environment — this prices production only.
- Vendor DevOps time to build and maintain this — a separate team-cost line, priced elsewhere in this project's costing.

THREE NUMBERS HERE ARE ASSUMPTIONS, NOT MEASUREMENTS — CONFIRM BEFORE SANCTION

1. Usage per student is estimated, not observed. 1.5 uploads/day at 3 MB and 3 re-views per document are reasonable starting assumptions, not telemetry. If real usage runs double this, the bill moves from **₹10.28 L to roughly ₹16–17 L** — bandwidth and storage are what move, not the servers. **2. This prices AWS Mumbai as the benchmark**, not a confirmed hosting decision. If this ultimately runs on NIC's MeghRaj government cloud, get their quote before sanction — MeghRaj doesn't publish a comparable public rate card, so this figure is the honest commercial-market baseline to negotiate against. **3. The 1-year Reserved discount (~30%) is a market-typical estimate**, not a quoted rate — confirm the exact figure in the AWS Pricing Calculator before committing to a term, since breaking a Reserved commitment early forfeits the discount.